

# Models for Count Outcomes

Poisson regression and variations

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## 1 Models for Count Outcomes

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### Acknowledgements

This content is adapted from:

- Dobson and Barnett (2018), Chapter 9
  - Vittinghoff et al. (2012), Chapter 8
- 
- 

### Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
```

```

library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times

```

Here are some R settings I use in this document:

```

rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE

```

## 1.1 Introduction

This chapter presents models for count data<sup>1</sup> outcomes. With covariates, the event rate  $\lambda$  becomes a function of the covariates  $\tilde{X} = (X_1, \dots, X_n)$ . Typically, count data models use a  $\log\{\}$  link function, and thus an  $\exp\{\}$  inverse-link function. That is:

$$\begin{aligned}
 E[Y|\tilde{X} = \tilde{x}, T = t] &= \mu(\tilde{x}, t) \\
 \mu(\tilde{x}, t) &= \lambda(\tilde{x}) \cdot t \\
 \lambda(\tilde{x}) &= \exp\{\eta(\tilde{x})\} \\
 \eta(\tilde{x}) &= \tilde{x}'\tilde{\beta} = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p
 \end{aligned} \tag{1}$$

---

<sup>1</sup>[probability.qmd#sec-count-vars](#)

$T = t$  is called the *exposure magnitude* (**?@def-exposure**) and has a special role in this model.

---

**Exercise 1.1.** Where have we seen a relationship like

$$\mu = \lambda \cdot t$$

before?

---

*Solution 1.1.* The relationship

$$\mu = \lambda \cdot t$$

in count regression models is analogous to the relationship

$$\mu = n\pi$$

in Binomial models.

---

We can also think of  $t$  as a special part of the linear component:

$$\begin{aligned}\log\{E[Y|\tilde{X} = \tilde{x}, T = t]\} &= \log\{\mu(\tilde{x})\} \\ &= \log\{\lambda(\tilde{x}) \cdot t\} \\ &= \log\{\lambda(\tilde{x})\} + \log t \\ &= \log\{\exp\{\eta(\tilde{x})\}\} + \log t \\ &= \eta(\tilde{x}) + \log t \\ &= \tilde{x}'\tilde{\beta} + \log t \\ &= (\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p) + \log t\end{aligned}$$

In contrast with the other covariates (represented by  $\tilde{X}$ ),  $t$  enters this expression with a log transformation and without a corresponding  $\beta$  coefficient; in other words,  $\log\{t\}$  is an **offset term** (**?@def-offset**).

---

**Exercise 1.2.** What are the units of  $\mu$  in Equation 1?

---

*Solution 1.2.*  $\mu$  is the mean of  $Y$ , and  $Y$  is a count, so  $\mu$  is also a count; for example:

- 3.1 cyclones,
- 10.23 ER visits
- 15.01 infections

---

**Exercise 1.3.** What are the units of  $\lambda$  in Equation 1?

---

*Solution 1.3.*  $\lambda = \mu/t$ , so  $\lambda$  is a rate of counts per unit of  $t$ . For example:

- 3.1 cyclones *per year*
- 2.023 ER visits per 10 person-years
- 15.01 infections per 1000 person-years at risk

## 1.2 Interpreting Poisson regression models

Differences on the log-rate scale become ratios on the rate scale, because

$$\exp\{a - b\} = \frac{\exp\{a\}}{\exp\{b\}}$$

(recall from Algebra 2<sup>2</sup>)

Therefore, according to this model, **differences of  $\delta$  in covariate  $x_j$  correspond to rate ratios of  $\exp\{\beta_j \cdot \delta\}$ .**

That is, letting  $\tilde{X}_{-j}$  denote vector  $\tilde{X}$  with element  $j$  removed:

$$\begin{aligned} & \left\{ \begin{array}{l} \log E[Y | X_j = a, \tilde{X}_{-j} = \tilde{x}_{-j}, T = t] \\ -\log E[Y | X_j = b, \tilde{X}_{-j} = \tilde{x}_{-j}, T = t] \end{array} \right\} \\ &= \left\{ \begin{array}{l} \log t + \beta_0 + \beta_1 x_1 + \dots + \beta_j(a) + \dots + \beta_p x_p \\ -\log t + \beta_0 + \beta_1 x_1 + \dots + \beta_j(b) + \dots + \beta_p x_p \end{array} \right\} \\ &= \beta_j(a - b) \end{aligned}$$

And accordingly,

$$\frac{E[Y | X_j = a, \tilde{X}_{-j} = \tilde{x}_{-j}, T = t]}{E[Y | X_j = b, \tilde{X}_{-j} = \tilde{x}_{-j}, T = t]} = \exp\{\beta_j(a - b)\}$$

## 1.3 Example: needle-sharing

(adapted from Vittinghoff et al. (2012), §8)

---

```
library(tidyverse)
library(haven)
needles =
  "inst/extdata/needle_sharing.dta" |>
  read_dta() |>
  as_tibble() |>
  mutate(
    hivstat =
      hivstat |>
      case_match(
        1 ~ "HIV+",
        0 ~ "HIV-") |>
      factor() |>
      relevel(ref = "HIV-"),
    polydrug =
      polydrug |>
      case_match(
        1 ~ "multiple drugs used",
        0 ~ "one drug used") |>
      factor() |>
      relevel(ref = "one drug used"),
    homeless =
      homeless |>
      case_match(
```

---

<sup>2</sup>[math-prereqs.qmd#cor-exp-sum](#)

```

    1 ~ "homeless",
    0 ~ "not homeless") |>
  factor() |>
  relevel(ref = "not homeless"),
  ethn = ethn |> factor() |> relevel(ref = "White"),
  sex = sex |> factor() |> relevel(ref = "M")
) |>
labelled::set_variable_labels(
  "sex" = "sex (reference = Male)",
  "ethn" = "ethnicity (reference = White)",
  "shsyryn" = "shared syringe yes/no (1 = yes, 0 = no)",
  "logshsyr" = "log(No. of shared needles)",
  "polydrug" = "how many drugs used?",
  "sqrtninj" = "sqrt(No. of infections in 30 days)",
  "homeless" = "Homeless (1 = yes, 0 = no)",
  "hivstat" = "HIV status (reference = HIV-)"
)

dict <- tibble(
  variable = names(needles),
  description = labelled::get_variable_labels(needles) |>
    sapply(function(x) ifelse(is.null(x), "", x)),
)
dict |> pander::pander()

```

Table 1: Data dictionary for **needles** data

| variable   | description                             |
|------------|-----------------------------------------|
| id         | ID                                      |
| sex        | sex (reference = Male)                  |
| ethn       | ethnicity (reference = White)           |
| age        | Age at 1st interview                    |
| dprsn_dx   | DPRSN_DX                                |
| sexabuse   | Sexually abused?                        |
| shared_syr | Shared syringe                          |
| hivstat    | HIV status (reference = HIV-)           |
| hplsns     | HPLSNS                                  |
| nivdu      | No of injections (in 30 days)           |
| shsyryn    | shared syringe yes/no (1 = yes, 0 = no) |
| sqrtnivd   | sqrt(No ivdu 30 days)                   |
| logshsyr   | log(No. of shared needles)              |
| polydrug   | how many drugs used?                    |
| sqrtninj   | sqrt(No. of infections in 30 days)      |
| homeless   | Homeless (1 = yes, 0 = no)              |
| shsyr      | No. of shared needles                   |

```

library(ggplot2)

needles |>
  ggplot(
    aes(
      x = age,
      y = shsyryn,
      shape = sex,

```

Table 2: Needle-sharing data

```

needles
#> # A tibble: 128 x 17
#>   id sex  ethn age dprsn_dx sexabuse shared_syr hivstat hplsns nivdu
#>   <dbl> <fct> <fct>   <dbl>   <dbl>   <dbl>   <dbl> <fct>   <dbl> <dbl>
#> 1 2104 M   White    47     5     0     1 HIV-     6    90
#> 2 2009 M   White    39     1     0     1 HIV+     2     4
#> 3 2032 M   White    52     1     0     1 HIV-    18    90
#> 4 2063 M   AA      47     1     1     1 HIV-     1   120
#> 5 2059 M   Hispanic 32     1     0     2 HIV-    12   120
#> 6 2077 M   Hispanic 54     1     0     2 HIV-    10   120
#> 7 2042 F   White    32     5     0     2 HIV-     8    15
#> 8 2017 M   White    26     5     0     2 HIV-    11   120
#> 9 2119 M   White    54     1     0     3 HIV-     2    90
#> 10 2085 F   White    19     5     0     3 HIV-     7    90
#> # i 118 more rows
#> # i 7 more variables: shsyryn <dbl>, sqrtnivd <dbl>, logshsyr <dbl>,
#> #   polydrug <fct>, sqrtninj <dbl>, homeless <fct>, shsyr <dbl>

```

```

    col = ethn
  )
) +
geom_point(
  aes(size = nivdu),
  alpha = .5) +
scale_size_area(max_size = 4) +
facet_grid(
  cols = vars(polydrug),
  rows = vars(homeless)) +
theme(legend.position = "bottom")

```

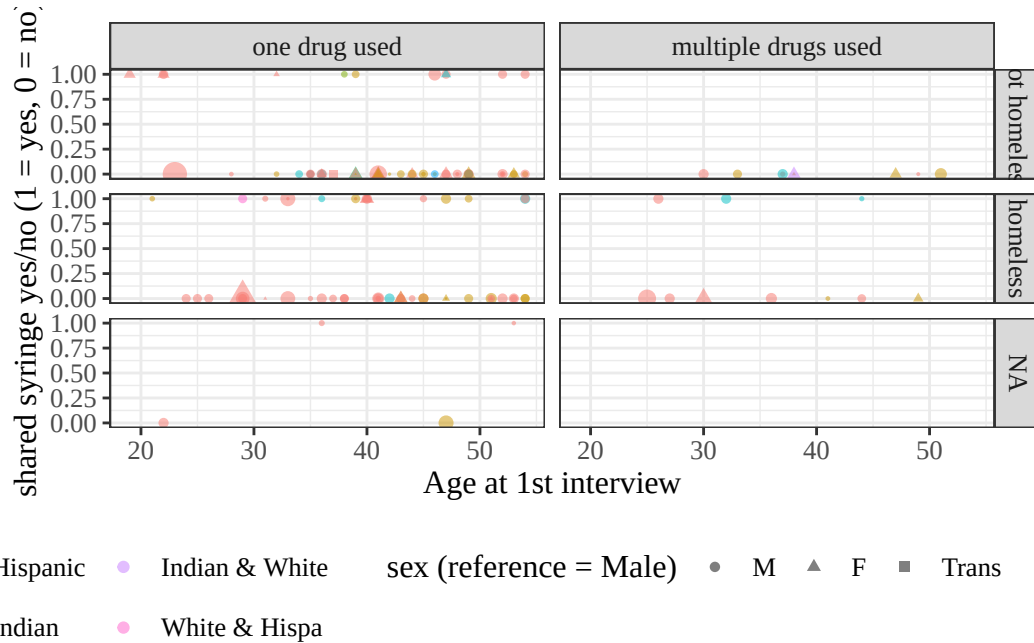


Figure 1: Rates of needle sharing

Table 3: Counts of observations in `needles` dataset by sex, unhoused status, and multiple drug use

```
needles |>
  dplyr::select(sex, homeless, polydrug) |>
  summary()
#>      sex      homeless      polydrug
#> M      :97    not homeless:63    one drug used      :109
#> F      :30    homeless   :61    multiple drugs used: 19
#> Trans: 1     NA's       : 4
```

## Covariate counts

There's only one individual with `sex = Trans`, which unfortunately isn't enough data to analyze. We will remove that individual:

```
needles = needles |> filter(sex != "Trans")
```

### 1.3.1 Model

```
glm1 =
  needles |>
  dplyr::filter(nivdu > 0) |>
  glm(
    offset = log(nivdu),
    family = stats::poisson,
    formula = shared_syr ~ age + sex + homeless*polydrug
  )
library(equationomatic)
equationomatic::extract_eq(glm1)
```

$$\log(E(\text{shared}_{\text{syr}})) = \alpha + \beta_1(\text{age}) + \beta_2(\text{sex}_F) + \beta_3(\text{homeless}) + \beta_4(\text{polydrug}) + \beta_5(\text{homeless} \times \text{polydrug}) + (\text{offset}) \quad (2)$$

```
library(parameters)
glm1 |> parameters(exponentiate = TRUE) |>
  print_md()
```

Table 4: Poisson model for needle-sharing data

| Parameter                                            | IRR      | SE       | 95% CI                 | z      | p      |
|------------------------------------------------------|----------|----------|------------------------|--------|--------|
| (Intercept)                                          | 0.02     | 4.02e-03 | (9.09e-03, 0.03)       | -15.87 | < .001 |
| age                                                  | 1.00     | 5.85e-03 | (0.99, 1.01)           | 0.42   | 0.673  |
| sex (F)                                              | 1.37     | 0.16     | (1.09, 1.72)           | 2.73   | 0.006  |
| homeless (homeless)                                  | 2.77     | 0.34     | (2.18, 3.53)           | 8.30   | < .001 |
| polydrug (multiple drugs used)                       | 2.85e-07 | 8.71e-05 | (1.59e-267, 5.11e+253) | -0.05  | 0.961  |
| homeless (homeless) × polydrug (multiple drugs used) | 6.21e+05 | 1.90e+08 | (3.47e-255, 1.11e+266) | 0.04   | 0.965  |

```
library(sjPlot)
glm1 |>
  sjPlot::plot_model(
    type = "pred",
    terms = c("age", "sex", "homeless", "polydrug"),
    show.data = TRUE
  ) +
  theme(legend.position = "bottom")
```

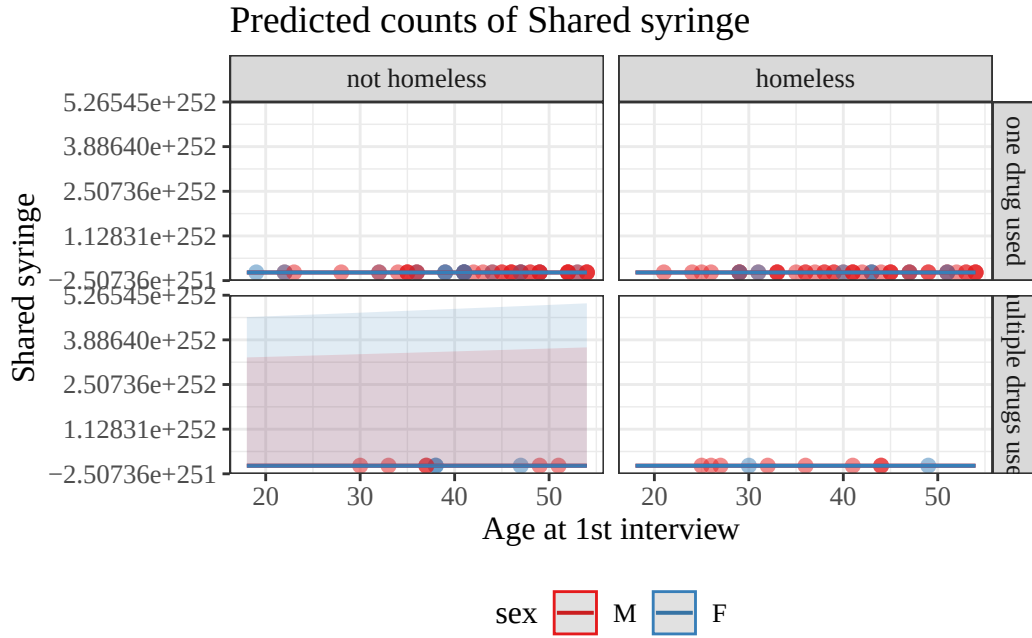


Figure 2

## 1.4 Inference for count regression models

### 1.4.1 Confidence intervals for regression coefficients and rate ratios

As usual:

$$\beta \in [\hat{\beta} \pm z_{1-\frac{\alpha}{2}} \cdot \widehat{\text{se}}(\hat{\beta})]$$

Rate ratios: exponentiate CI endpoints

$$\exp\{\beta\} \in [\exp\{\hat{\beta} \pm z_{1-\frac{\alpha}{2}} \cdot \widehat{\text{se}}(\hat{\beta})\}]$$

### 1.4.2 Hypothesis tests for regression coefficients

$$z = \frac{\hat{\beta} - \beta_0}{\widehat{\text{se}}(\hat{\beta})}$$

Compare  $z$  or  $|z|$  to the tails of the standard Gaussian distribution, according to the null hypothesis.

### 1.4.3 Comparing nested models

log(likelihood ratio) tests, as usual.

## 1.5 Prediction

$$\begin{aligned}\hat{y} &\stackrel{\text{def}}{=} \hat{\text{E}}[Y|\tilde{X} = \tilde{x}, T = t] \\ &= \hat{\mu}(\tilde{x}, t) \\ &= \hat{\lambda}(\tilde{x}) \cdot t \\ &= \exp\{\hat{\eta}(\tilde{x})\} \cdot t \\ &= \exp\left\{\tilde{x}'\hat{\beta}\right\} \cdot t\end{aligned}$$

## 1.6 Diagnostics

### 1.6.1 Residuals

Observation residuals

$$e \stackrel{\text{def}}{=} y - \hat{y}$$

Pearson residuals

$$r = \frac{e}{\widehat{\text{se}}(e)} \approx \frac{e}{\sqrt{\hat{y}}}$$

Standardized Pearson residuals

$$r_p = \frac{r}{\sqrt{1-h}}$$

where  $h$  is the “leverage” (which we will continue to leave undefined).

Deviance residuals

$$d_k = \text{sign}(y - \hat{y}) \left\{ \sqrt{2[\ell_{\text{full}}(y) - \ell(\hat{\beta}; y)]} \right\}$$

**i** Note

$$\text{sign}(x) \stackrel{\text{def}}{=} \frac{x}{|x|}$$

In other words:

- $\text{sign}(x) = -1$  if  $x < 0$
- $\text{sign}(x) = 0$  if  $x = 0$
- $\text{sign}(x) = 1$  if  $x > 0$

```
library(ggfortify)
autoplot(glm1)
```

## 1.7 Zero-inflation

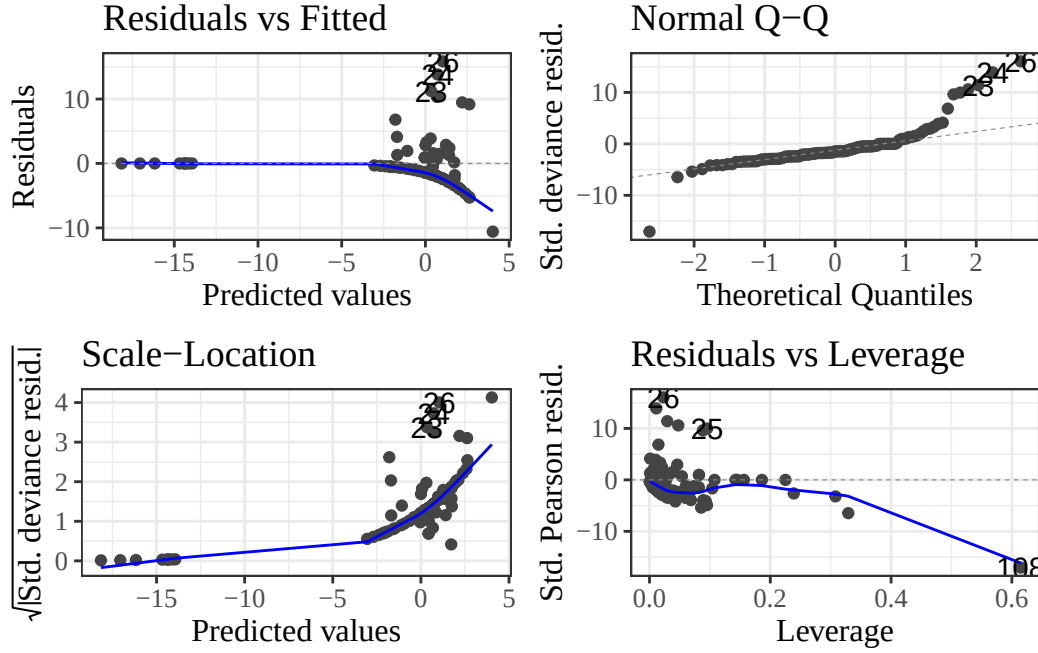
### 1.7.1 Models for zero-inflated counts

We assume a latent (unobserved) binary variable,  $Z$ , which we model using logistic regression:

$$P(Z = 1|X = x) = \pi(x) = \text{expit}(\gamma_0 + \gamma_1 x_1 + \dots)$$

According to this model, if  $Z = 1$ , then  $Y$  will always be zero, regardless of  $X$  and  $T$ :

Table 5: Diagnostics for Poisson model



$$P(Y = 0|Z = 1, X = x, T = t) = 1$$

Otherwise (if  $Z = 0$ ),  $Y$  will have a Poisson distribution, conditional on  $X$  and  $T$ , as above.

Even though we never observe  $Z$ , we can estimate the parameters  $\gamma_0$ - $\gamma_p$ , via maximum likelihood:

$$P(Y = y|X = x, T = t) = P(Y = y, Z = 1|...) + P(Y = y, Z = 0|...)$$

(by the Law of Total Probability)

where

$$P(Y = y, Z = z|...) = P(Y = y|Z = z, ...)P(Z = z|...)$$

---

**Exercise 1.4.** Expand  $P(Y = 0|X = x, T = t)$ ,  $P(Y = 1|X = x, T = t)$  and  $P(Y = y|X = x, T = t)$  into expressions involving  $P(Z = 1|X = x, T = t)$  and  $P(Y = y|Z = 0, X = x, T = t)$ .

---

**Exercise 1.5.** Derive the expected value and variance of  $Y$ , conditional on  $X$  and  $T$ , as functions of  $P(Z = 1|X = x, T = t)$  and  $E[Y|Z = 0, X = x, T = t]$ .

## 1.8 Over-dispersion

---

The Poisson distribution model **forces** the variance to equal the mean. In practice, many count distributions will have a variance substantially larger than the mean (or occasionally, smaller).

**Definition 1.1** (Overdispersion). A random variable  $X$  is **overdispersed** relative to a model  $p(X = x)$  if its empirical variance in a dataset is larger than the value is predicted by the fitted model  $\hat{p}(X = x)$ .

c.f. Dobson and Barnett (2018) §3.2.1, 7.7, 9.8; Vittinghoff et al. (2012) §8.1.5; and <https://en.wikipedia.org/wiki/Overdispersion>.

When we encounter overdispersion, we can try to reduce the residual variance by adding more covariates.

---

**i** Note

Logistic regression is named after the (inverse) link function. Poisson regression is named after the outcome distribution. I think this naming convention reflects the strongest (most questionable assumption) in the model. In binary data regression, the outcome distribution essentially *must* be Bernoulli (or Binomial), but the link function could be logit, log, identity, probit, or something more unusual. In count data regression, the outcome distribution could have many different shapes, but the link function will probably end up being log, so that covariates have multiplicative effects on the rate.

---

### 1.8.1 Negative binomial models

There are alternatives to the Poisson model. Most notably, the negative binomial model<sup>3</sup>.

We can still model  $\mu$  as a function of  $X$  and  $T$  as before, and we can combine this model with zero-inflation (as the conditional distribution for the non-zero component).

---

#### Example: needle-sharing

```
library(MASS) #need this for glm.nb()
glm1.nb = glm.nb(
  data = needles,
  shared_syr ~ age + sex + homeless*polydrug
)

equationomatic::extract_eq(glm1.nb)
```

$$\log(E(\text{shared}_{\text{syr}})) = \alpha + \beta_1(\text{age}) + \beta_2(\text{sex}_F) + \beta_3(\text{homeless}) + \beta_4(\text{polydrug}) + \beta_5(\text{homeless} \times \text{polydrug}) \quad (3)$$

---

#### zero-inflation

```
library(glmmTMB)
zinf_fit1 = glmmTMB(
  family = "poisson",
  data = needles,
  formula = shared_syr ~ age + sex + homeless*polydrug,
  ziformula = ~ age + sex + homeless + polydrug # fit won't converge with interaction
)

zinf_fit1 |>
  parameters(exponentiate = TRUE) |>
  print_md()
```

---

<sup>3</sup>[probability.qmd#sec-nb-dist](#)

Table 6: Negative binomial model for needle-sharing data

```
summary(glm1.nb)
#>
#> Call:
#> glm.nb(formula = shared_syr ~ age + sex + homeless * polydrug,
#> data = needles, init.theta = 0.08436295825, link = log)
#>
#> Coefficients:
#>
#> Estimate Std. Error z value
#> (Intercept) 9.91e-01 1.71e+00 0.58
#> age -2.76e-02 3.82e-02 -0.72
#> sexF 1.06e+00 8.07e-01 1.32
#> homelesshomeless 1.65e+00 7.22e-01 2.29
#> polydrugmultiple drugs used -2.46e+01 3.61e+04 0.00
#> homelesshomeless:polydrugmultiple drugs used 2.32e+01 3.61e+04 0.00
#>
#> Pr(>|z|)
#> (Intercept) 0.563
#> age 0.469
#> sexF 0.187
#> homelesshomeless 0.022 *
#> polydrugmultiple drugs used 0.999
#> homelesshomeless:polydrugmultiple drugs used 0.999
#> ---
#> Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> (Dispersion parameter for Negative Binomial(0.0844) family taken to be 1)
#>
#> Null deviance: 69.193 on 119 degrees of freedom
#> Residual deviance: 57.782 on 114 degrees of freedom
#> (7 observations deleted due to missingness)
#> AIC: 315.5
#>
#> Number of Fisher Scoring iterations: 1
#>
#>
#> Theta: 0.0844
#> Std. Err.: 0.0197
#>
#> 2 x log-likelihood: -301.5060
```

Table 7: Poisson versus Negative Binomial Regression coefficient estimates

```
tibble(name = names(coef(glm1)), poisson = coef(glm1), nb = coef(glm1.nb))
#> # A tibble: 6 x 3
#> name poisson nb
#> <chr> <dbl> <dbl>
#> 1 (Intercept) -4.18 0.991
#> 2 age 0.00247 -0.0276
#> 3 sexF 0.316 1.06
#> 4 homelesshomeless 1.02 1.65
#> 5 polydrugmultiple drugs used -15.1 -24.6
#> 6 homelesshomeless:polydrugmultiple drugs used 13.3 23.2
```

Table 8: Zero-inflated poisson model

| Table 8: # Fixed Effects                             |          |          |              |           |        |
|------------------------------------------------------|----------|----------|--------------|-----------|--------|
| Parameter                                            | IRR      | SE       | 95% CI       | z         | p      |
| (Intercept)                                          | 3.16     | 0.82     | (1.90, 5.25) | 4.44      | < .001 |
| age                                                  | 1.01     | 5.88e-03 | (1.00, 1.02) | 1.74      | 0.081  |
| sex [F]                                              | 3.43     | 0.44     | (2.67, 4.40) | 9.68      | < .001 |
| homeless [homeless]                                  | 3.44     | 0.47     | (2.63, 4.50) | 9.03      | < .001 |
| polydrug [multiple drugs used]                       | 1.85e-09 | 1.21e-05 | (0.00, Inf)  | -3.08e-03 | 0.998  |
| homeless [homeless] × polydrug [multiple drugs used] | 1.38e+08 | 9.04e+11 | (0.00, Inf)  | 2.87e-03  | 0.998  |

Table 9: Zero-inflated poisson model

| Table 9: # Zero-Inflation      |            |      |              |       |       |
|--------------------------------|------------|------|--------------|-------|-------|
| Parameter                      | Odds Ratio | SE   | 95% CI       | z     | p     |
| (Intercept)                    | 0.49       | 0.54 | (0.06, 4.25) | -0.65 | 0.514 |
| age                            | 1.05       | 0.03 | (1.00, 1.10) | 1.95  | 0.051 |
| sex [F]                        | 1.44       | 0.84 | (0.46, 4.50) | 0.62  | 0.533 |
| homeless [homeless]            | 0.68       | 0.34 | (0.26, 1.80) | -0.78 | 0.436 |
| polydrug [multiple drugs used] | 1.15       | 0.91 | (0.24, 5.43) | 0.18  | 0.858 |

Another R package for zero-inflated models is `pscl`<sup>4</sup> (Zeileis et al. (2008)).

#### zero-inflated negative binomial model

```
library(glmTMB)
zinf_fit1 = glmTMB(
  family = nbinom2,
  data = needles,
  formula = shared_syr ~ age + sex + homeless*polydrug,
  ziformula = ~ age + sex + homeless + polydrug
  # fit won't converge with interaction
)

zinf_fit1 |>
  parameters(exponentiate = TRUE) |>
  print_md()
```

Table 10: Zero-inflated negative binomial model

| Table 10: # Fixed Effects |      |      |               |      |       |
|---------------------------|------|------|---------------|------|-------|
| Parameter                 | IRR  | SE   | 95% CI        | z    | p     |
| (Intercept)               | 1.06 | 1.48 | (0.07, 16.52) | 0.04 | 0.969 |
| age                       | 1.02 | 0.03 | (0.96, 1.08)  | 0.53 | 0.599 |

<sup>4</sup><https://cran.r-project.org/web/packages/pscl/index.html>

| Parameter                                            | IRR      | SE       | 95% CI        | z         | p     |
|------------------------------------------------------|----------|----------|---------------|-----------|-------|
| sex [F]                                              | 6.86     | 6.36     | (1.12, 42.16) | 2.08      | 0.038 |
| homeless [homeless]                                  | 6.44     | 4.59     | (1.60, 26.01) | 2.62      | 0.009 |
| polydrug [multiple drugs used]                       | 8.25e-10 | 7.07e-06 | (0.00, Inf)   | -2.44e-03 | 0.998 |
| homeless [homeless] × polydrug [multiple drugs used] | 2.36e+08 | 2.02e+12 | (0.00, Inf)   | 2.25e-03  | 0.998 |

Table 11: Zero-inflated negative binomial model

| Table 11: # Zero-Inflation     |            |      |                  |       |       |
|--------------------------------|------------|------|------------------|-------|-------|
| Parameter                      | Odds Ratio | SE   | 95% CI           | z     | p     |
| (Intercept)                    | 0.10       | 0.20 | (1.47e-03, 6.14) | -1.11 | 0.269 |
| age                            | 1.07       | 0.04 | (0.99, 1.15)     | 1.78  | 0.075 |
| sex [F]                        | 2.72       | 2.40 | (0.48, 15.33)    | 1.13  | 0.258 |
| homeless [homeless]            | 1.15       | 0.86 | (0.27, 4.96)     | 0.19  | 0.853 |
| polydrug [multiple drugs used] | 0.75       | 0.86 | (0.08, 7.12)     | -0.25 | 0.799 |

Table 12: Zero-inflated negative binomial model

| Table 12: # Dispersion |             |              |
|------------------------|-------------|--------------|
| Parameter              | Coefficient | 95% CI       |
| (Intercept)            | 0.44        | (0.11, 1.71) |

### 1.8.2 Quasipoisson

An alternative to Negative binomial is the “quasipoisson” distribution. I’ve never used it, but it seems to be a method-of-moments type approach rather than maximum likelihood. It models the variance as  $\text{Var}(Y) = \mu\theta$ , and estimates  $\theta$  accordingly.

See `?quasipoisson` in R for more.

## 1.9 More on count regression

- <https://bookdown.org/roback/bookdown-BeyondMLR/ch-poissonreg.html>

Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. <https://doi.org/10.1201/9781315182780>.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.

Zeileis, Achim, Christian Kleiber, and Simon Jackman. 2008. “Regression Models for Count Data in R.” *Journal of Statistical Software* 27 (8). <https://www.jstatsoft.org/v27/i08/>.